

RAJAT BISHT

SOFTWARE | DATA SCIENCE | AI & MACHINE LEARNING

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EXPERIENCE

Data Science Intern . Celebal Technologies

June 2026 - Present

CSI 2026 Program . Remote

- Developed and evaluated regression and time-series forecasting models (ARIMA) on automotive pricing data, applying cross-validation and GridSearchCV for hyperparameter optimisation.
- Built end-to-end ML pipelines across supervised learning, clustering, and deep learning tasks (ANN vs CNN on CIFAR-10), with model evaluation using precision, recall, F1-score, and ROC-AUC.

PROJECTS

AI-Driven Fraud Detection System github.com/rb937/fraud-detection-system

Nov 2025 - Jan 2026

Python . Scikit-Learn . Pandas . Streamlit

- Engineered a real-time fraud detection pipeline using Random Forest, achieving **92% Recall** on a heavily imbalanced dataset via SMOTE oversampling and cost-sensitive learning.
- Developed geospatial risk features using Haversine Distance to flag location anomalies (transactions >500 km from home); deployed an interactive Streamlit dashboard for real-time inference.

Cortex . Local RAG System github.com/rb937/Cortex

Jun 2026 - Jul 2026

Python . ChromaDB . Ollama . LLMs . RAG

- Built a local-first RAG system indexing **500+ pages** across different documents with 500-token chunks and 50-token overlap into a ChromaDB vector store, achieving sub-minute full corpus indexing, accurate cross-document retrieval with source citation, and graceful out-of-corpus handling.
- Designed a swappable backend interface supporting Ollama (local), OpenAI, and Anthropic, enabling model-agnostic deployment across local and cloud LLM providers with zero code changes.

ChromaEngine . Image Processing Engine github.com/rb937/ChromaEngine

Dec 2025 - Feb 2026

Java . JavaFX . AWT . Algorithms . OOP

- Engineered a non-destructive image processing engine in Java/JavaFX handling 4K resolution input, implementing 5 real-time filters with live preview using direct-memory bitwise operations on ARGB channels, bypassing BufferedImage overhead entirely.
- Implemented custom 3x3 convolution matrices (sharpen, blur, edge detection) and a spatial transformation pipeline demonstrating applied linear algebra, matrix operations, and algorithmic optimisation.

SKILLS

Languages

Python, Java, C, R, SQL

ML

Scikit-Learn, Pandas, NumPy, Matplotlib, Seaborn

AI / LLM

ChromaDB, Ollama, LangChain, nomic-embed-text

Frameworks

Streamlit, JavaFX, AWT / Swing

Tools

Git, MS Excel, Power BI, Jupyter Notebook

EDUCATION

B.Tech Computer Science Engineering (Data Science)

2023 - 2027

DIT University, Dehradun . CGPA: 8.0 / 10

CERTIFICATIONS

Programming Fundamentals - Duke University

Database Management Essentials - University of Colorado

Data Fundamentals - IBM